

Dual Component Model of Mass Gravity and Spatial Twist Stress Gravity

A WRRRA Reinterpretation of the MOND Low Acceleration Relation and Its Cosmological Extrapolation

Wonsik Choi

Independent Researcher · Seoul, Republic of Korea

janefather@gmail.com

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Key Numerical Result

Using the present-universe normalization $H_0=67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_b=0.0493$, and $\Omega_T=0.265$, the cosmic mean equivalent source ratio between direct baryonic gravity and twist stress is $g_T:g_M \approx 5.37:1$. Within these two attractive components, twist stress contributes 84.3% and the direct baryonic component 15.7%. This ratio follows from interpreting the WRRRA dark component as spatial stress; by itself it does not establish the existence of twist stress.

Research Status of This Paper

This paper is a self-contained research note whose definitions and calculations can be reproduced from the material presented here. The public preprint DOI 10.5281/zenodo.22700557 provides the earlier form of the twist-stress and galaxy-acceleration connection. DOI 10.5281/zenodo.22931512 provides the horizon Renderer, the area-bit calibration, the competing gravity branches, and the rule against double counting. Both works are cited to document the research genealogy and the origin of definitions, while all numerical calculations used here are restated in this paper. Values fixed by established constants or observations are classified as verified inputs or calibrated closure relations; outputs in unmeasured domains obtained after calibration are classified as predictions.

Abstract

This study decomposes the gravitational field observed in WRRRA into direct mass gravity g_M produced by baryonic matter and an additional component g_T produced by stress stored in spatial twist and deformation. The decomposition is not a claim that two independent fundamental forces exist. It is a weak-field Renderer bookkeeping device that distinguishes the direct material source from the spatial response. The MOND low-acceleration relation $g \approx \sqrt{a_T g_M}$ is rewritten so that the constitutive response $\chi_T(g_M/a_T)$ grows at low acceleration, instead of asserting an arbitrary change of physical law in a selected regime. With the present-universe calibration $H_0=67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $f_T=0.265$, and the input

constant c , the closure relation $a_T=cH_0\sqrt{(f_T/8)}$ gives $a_T=1.1918\times10^{-10} \text{ m s}^{-2}$. This differs by about 0.75% from the diagnostic value $1.2007\times10^{-10} \text{ m s}^{-2}$ obtained from the low-acceleration SPARC data. Converting an observable-universe radius of 46.3 billion light-years into a uniform spherical equivalent source gives $g_M\approx5.15\times10^{-11} \text{ m s}^{-2}$ and $g_T\approx2.77\times10^{-10} \text{ m s}^{-2}$, with a ratio of 5.37:1. These absolute accelerations are Newtonian equivalent quantities for comparison, not actual global forces in an expanding universe. The model therefore passes a first reality-consistency test. The causal hypothesis of twist stress will require the same spatial response function to account simultaneously for galaxy rotation, lensing, galaxy clusters, and cosmic structure growth.

Keywords WRRa · mass gravity · spatial twist stress · MOND · SPARC · cosmic mean source ratio

Core Assessment

Category	Current assessment
Verified input	c , G , H_0 , Ω_b , and SPARC observations
WRRa-specific transformation	global twist fraction $\rightarrow a_T$ closure relation $\rightarrow$ spatial response χ_T
Output	$a_T=1.1918\times10^{-10} \text{ m s}^{-2}$; cosmic mean $g_T/g_M=5.37$
Falsification condition	one fixed response function fails to reproduce rotation, lensing, clusters, and growth simultaneously
Conclusion	first reality-consistency test passed; twist causation and galaxy-by-galaxy universality remain unresolved

1. Research Purpose and Question

In the standard weak-field description, gravity follows the mass distribution of baryonic matter. In galaxy outskirts, however, the observed acceleration exceeds the Newtonian acceleration calculated from visible baryons alone. This discrepancy has been described either by adding unseen matter or by modifying the dynamical response law at low acceleration.

WRRR asks a different question. Can the direct gravity generated by mass be separated from gravity generated by twist stress that the mass and the state of the universe leave in space? The latter may be negligible for ordinary masses at high acceleration, yet become relatively important at low acceleration, large radius, and low surface density.

$$g_{total} = g_M + g_T$$

two-component bookkeeping in the weak-field Renderer

(1)

In Eq. (1), g_M is the direct component produced by baryonic matter, and g_T is the additional component produced by stress stored in spatial twist and deformation. The term g_T does not presuppose particle dark matter. It is a candidate representation of a spatial response that appears observationally as dark mass.

Important Limitation

It is incorrect to state that the relative twist effect must grow whenever mass grows. The absolute twist load may increase with mass, but its ratio to the direct component is determined by local acceleration, radius, surface density, and the response function rather than by total mass alone.

2. Scope and Assessment Rules

This study uses a toy model of our universe fixed to established constants. The speed of light c is not predicted; it is an input that fixes the Renderer conversion between length and time. H_0 and the cosmic component fractions are also observational inputs. If the resulting closure relation reproduces the known galaxy-acceleration scale, the model is connected to reality. Once the closure relation and response function are frozen, their outputs for other phenomena become WRRR predictions.

Research status	Meaning
Definition	decomposition into g_M and g_T ; relation between χ_T and v
Derived result	r_T , M_T , and ρ_T obtained algebraically from the fixed equations
Calibrated closure relation	$a_T = c H_0 v (f_T/8)$; the factor 1/8 is a cross-scale connection calibrated to the present universe
External support	numerical agreement with the SPARC RAR and cosmic component fractions
Hypothesis	interpretation of spatial twist stress as the physical cause of dark-mass phenomena
Prediction	unused outputs for lensing, clusters, redshift evolution, and growth after calibration is frozen

3. Present Universe Normalization

This paper fixes three WRRa premises: minimal computation, a common carrier, and phenotype. All physical definitions and numerical values required for the calculation are restated below without relying on an internal document. Observational inputs for the present universe are then used to connect the low-acceleration scale of twist stress through one closure relation.

An earlier public form of this cross-scale connection appeared in Choi's Zenodo preprint [1]. A subsequent horizon study distinguished competing branches, including B_Q, which treats twist stress as an additional effective source, and B_C, which treats it as a geometric response. That study also required the branches not to be added twice in one calculation [2]. Building on this lineage, the present paper reconstructs the decomposition of mass gravity and two-dimensional twist stress, the surface-density threshold, the directional feedback of disks, and the cosmic mean component ratio.

$$a_T = c H_0 \sqrt{(f_T / 8)} \quad (2)$$

c is an input and is not predicted

Symbol	Value	Status
c	299 792 458 m s ⁻¹	defined input
G	6.67430×10 ⁻¹¹ m ³ kg ⁻¹ s ⁻²	Verified input
H ₀	67.4 km s ⁻¹ Mpc ⁻¹	present-universe calibration input
f _T	0.265	twist-equivalent component fraction
a _T	1.191812669×10 ⁻¹⁰ m s ⁻²	output of Eq. (2)

The critical density and twist-equivalent density are

$$\rho_c = 3 H_0^2 / (8 \pi G) = 8.533 \times 10^{-27} \text{ kg m}^{-3} \quad (3)$$

$$\rho_T = f_T \rho_c = 2.261 \times 10^{-27} \text{ kg m}^{-3} \quad (4)$$

$$u_T = \rho_T c^2 = 2.032 \times 10^{-10} \text{ J m}^{-3} \quad (5)$$

The background twist-stress energy density is defined here as $u_T = \zeta c^4 \kappa^2 / (16 \pi G)$. Combining this definition with Eq. (5) fixes only the product $\zeta \kappa^2 = 8.441 \times 10^{-53} \text{ m}^{-2}$. It does not constitute separate measurements of ζ and κ . The present background twist index $\mathcal{W}_0 = \sqrt{\zeta \kappa (c/H_0)} = \sqrt{(6f_T)} = 1.26095$ is likewise a normalization index, not a directly observed physical angle.

4. Definition of the Two Gravitational Components

In a spherically symmetric weak field, the potential Φ_M generated by baryonic density ρ_b is calculated first.

$$\nabla^2 \Phi_M = 4 \pi G \rho_b, \quad g_M = -\nabla \Phi_M \quad (6)$$

The spatial twist response is represented by the dimensionless function χ_T . The variable $y = g_M/a_T$ indicates whether direct gravity lies above or below the transition scale.

$$g_{-T} = \chi_{-T}(y) g_{-M}, \quad g = [1 + \chi_{-T}(y)] g_{-M} \quad (7)$$

$$\chi_{-T}(y) \rightarrow 0 \quad (y \gg 1), \quad \chi_{-T}(y) \rightarrow y^{-1/2} - 1 \quad (y \ll 1) \quad (8)$$

Writing $v(y)=1+\chi_{-T}(y)$ gives $v \rightarrow 1$ at high acceleration and $v \rightarrow y^{-1/2}$ at low acceleration. The physical content of the spatial stress is therefore encoded in the constitutive relation χ_{-T} or v . For WRRRA to go beyond a relabeling, this function must be fixed by an action or an equation of state for twist stress.

$$\nabla^2 \Phi = \nabla \cdot [v(|\nabla \Phi_{-M}|/a_{-T}) \nabla \Phi_{-M}]$$

QUMOND-type computational skeleton interpreted as a WRRRA spatial response

$$\rho_{-T}^{\text{eff}} = (1/4\pi G) \nabla \cdot [(v-1) \nabla \Phi_{-M}] \quad (10)$$

The quantity ρ_{-T}^{eff} in Eq. (10) is not a particle number density. It is the effective density obtained by inverting the same gravitational field as a Newtonian mass source, and therefore the mass phenotype of spatial twist stress.

5. WRRRA Reinterpretation of the MOND Relation

MOND captures the observed regularity at low acceleration in a compact form. Its phenomenological success must be retained. WRRRA differs by reading the same mathematical relation as a nonlinear spatial response rather than as an arbitrary modification of dynamics.

$$g \approx \sqrt{(a_{-T} g_{-M})} \quad (g_{-M} \ll a_{-T}) \quad (11)$$

$$g_{-T} = g - g_{-M} \approx \sqrt{(a_{-T} g_{-M})} - g_{-M} \quad (12)$$

Item	MOND phenomenology	WRRRA interpretation
Transition scale	a_0	$a_{-T} = c H_0 v(f_{-T}/8)$
Low-acceleration effect	change in effective dynamics or gravitational law	growth of the spatial twist response
Interpolation function	μ or v	constitutive relation χ_{-T} or v for spatial stress
Dark mass	not required	mass phenotype of stress represented by ρ_{-T}^{eff}
Open problem	microscopic cause	derive the response function from a twist action

This reinterpretation must be tested by computational closure, not by the novelty of its terminology. One a_{-T} and one $v(y)$ must be fixed and then tested against rotation curves, lensing, galaxy clusters, and cosmology without galaxy-by-galaxy retuning.

6. Quantitative Relation Between Mass and Transition Radius

For an isolated point mass M , $g_{-M} = GM/r^2$. The radius at which the direct component equals a_{-T} is

$$r_T = \sqrt{(GM/a_T)} \quad (13)$$

$$r_T \approx 0.0342 \sqrt{(M/M_\odot)} \text{ pc} \quad (14)$$

Mass M	Transition radius r_T	Interpretation
1 M _⊙	0.0342 pc ≈ 7 050 AU	outer stellar-system scale
10 ⁶ M _⊙	34.2 pc	star-cluster or compact-nucleus scale
10 ⁹ M _⊙	1.08 kpc	dwarf-galaxy scale
10 ¹⁰ M _⊙	3.42 kpc	inside a galactic disk
10 ¹¹ M _⊙	10.8 kpc	large-galaxy disk scale

Because r_T grows as $\sqrt{M}$, the absolute region in which the spatial response becomes important expands with mass. At a fixed radius, however, g_M is proportional to M , so g_T/g_M does not grow with total mass alone. A small low-surface-density galaxy may be more twist dominated than a larger galaxy.

Two Dimensional Twist and the Surface Density Threshold

The present hypothesis treats spatial twist as a two-dimensional deformation stress with direction and coherence over a broad area, rather than as a three-dimensional material density attached uniformly throughout a volume. Twist visibility should then depend more strongly on direct load per unit area than on total mass. Rewriting the point-mass field in terms of surface density exposes this structure.

$$\Sigma_b(r) = M/(\pi r^2), \quad g_M = \pi G \Sigma_b \quad (15)$$

$$\Sigma_{*}^{\wedge sph} = a_T/(\pi G) = 0.568 \text{ kg m}^{-2} \approx 272 \text{ M}_\odot \text{ pc}^{-2} \quad (16)$$

Normalizing the local field of a thin disk as $2\pi G \Sigma$ gives the corresponding threshold $\Sigma_{*}^{\wedge disk} = a_T/(2\pi G) \approx 0.284 \text{ kg m}^{-2} \approx 136 \text{ M}_\odot \text{ pc}^{-2}$. The factor-of-two difference comes from the geometric convention. Direct gravity dominates in compact high-density systems, whereas the relative effect of stress stored across a broad coherent two-dimensional area becomes visible when the same mass is spread to a larger radius and lower surface density.

This interpretation does not change the core of the toy model, but it identifies the active variable as the surface-density ratio equivalent to $y = g_M/a_T$. In the spherical approximation, $y = \Sigma_b/\Sigma_{*}^{\wedge sph}$. The effective condition for twist is therefore a sufficiently broad coherent area and a low surface-density threshold, not a large-total-mass threshold.

Additional Quantitative Condition

Two-dimensional geometry alone does not automatically produce a flat rotation curve. Obtaining $g_T \propto 1/r$ in the deep low-acceleration regime requires $M_T(r) \propto r$, so the equivalent two-dimensional stress surface density, or its gravitational projection, must vary approximately as $1/r$. Unless this scaling is derived from a twist action, the two-dimensional interpretation remains a descriptive analogy.

Disk Galaxies and Directional Feedback

The standard starting point is clear. As a three-dimensional gas cloud contracts, random vertical motion is dissipated by collisions and radiative cooling while total angular momentum is conserved. Contraction

reduces the moment of inertia, accelerates rotation, and produces a finite-thickness disk perpendicular to the remaining angular-momentum vector. Disk formation alone is therefore not evidence for WRRRA.

A flattened galaxy may strengthen two-dimensional twist stress, while directed twist stress may in turn collect three-dimensional matter more strongly into one plane of rotation. The present feedback hypothesis is initial angular momentum -> slight flattening -> stronger coherence of two-dimensional stress -> enhanced in-plane gravitational response -> a thinner and more stable disk.

A scalar response χ_T cannot select a plane. An anisotropic stress is required whose axial and disk-plane responses differ relative to the galactic angular-momentum axis $\hat{L}$.

$$\pi_{ij}^T = \Pi_T (\hat{L}_i \hat{L}_j - \delta_{ij}/3)$$

Rotation in the disk plane and vertical equilibrium test the same twist stress in different directions.

$$v_c^2/R = g_{M,R} + g_{T,R}$$

$$h_z \approx \sigma_z^2 / \{ \pi G [\Sigma_b + \Sigma_T^{\text{eff}}] \}$$

If Σ_T^{eff} inferred from the rotation curve predicts stellar vertical velocity dispersion σ_z and disk thickness h_z without retuning, twist stress would be supported as a contributor to disk formation and stability rather than only as a consequence of disk geometry. The directional hypothesis fails if g_T is isotropic or if the stress required by rotation predicts the wrong vertical structure.

The perpendicular rotation planes of polar-ring galaxies, outer-disk warps, the principal axes of elliptical galaxies, and the flattening of lensing mass distributions can discriminate the causal direction. Angular-momentum conservation and gas cooling remain the standard disk-formation mechanism. WRRRA must account quantitatively for any residual alignment of the additional gravitational field with the disk.

7. Effective Twist Mass and Density

In the spherical low-acceleration limit, converting the observed gravitational field into a Newtonian mass gives the effective mass of twist stress.

$$M_T(r) = r^2 [g - g_M] / G \quad (17)$$

$$M_T(r) \approx r \sqrt{(Ma_T/G)} - M \quad (18)$$

$$\rho_T(r) = \sqrt{(GMa_T)} / (4\pi Gr^2) \quad (19)$$

For $r \gg r_T$, the first term dominates, so M_T is approximately proportional to r and ρ_T decreases as r^{-2} . This is the characteristic slope of an equivalent halo that produces a flat rotation curve. Equation (19), however, is a deep-low-acceleration approximation for an isolated spherical system. A real disk requires solving the vector field in Eq. (9).

8. Extrapolation to the Observed Cosmic Scale

For the present universe, take $f_T=0.265$ as the spatial-stress representation of the dark component and use the Planck values $\Omega_b h^2=0.0224$ and $h=0.674$ for the direct baryonic component. This gives $\Omega_b=0.04931$. The present mean equivalent sources in a uniform spherical region are

$$M_i(R) = (4\pi/3)\rho_i R^3, \quad g_i^{eq}(R) = (4\pi G/3)\rho_i R \quad (20)$$

$$g_T^{eq}/g_M^{eq} = M_T/M_b = \rho_T/\rho_b = \Omega_T/\Omega_b = 5.374 \quad (21)$$

Component	Density parameter	Share of direct plus twist	Observable-universe equivalent mass
Direct baryonic component	$\Omega_b=0.04931$	15.69%	1.48×10^{53} kg
Twist-stress component	$\Omega_T=0.265$	84.31%	7.96×10^{53} kg
Ratio	$\Omega_T/\Omega_b=5.374$	5.37:1	5.37:1

Taking the present comoving radius of the observable universe as $R_{obs} \approx 46.3 \text{ Gly} \approx 4.38 \times 10^{26} \text{ m}$ gives the following comparison-only equivalent accelerations from Eq. (20).

Comparison radius	g_M^{eq}	g_T^{eq}	g_T^{eq}/g_M^{eq}
Hubble radius $c/H_0=14.51$ Gly	$1.61 \times 10^{-11} \text{ m s}^{-2}$	$8.68 \times 10^{-11} \text{ m s}^{-2}$	5.374
Observable-universe radius 46.3 Gly	$5.15 \times 10^{-11} \text{ m s}^{-2}$	$2.77 \times 10^{-10} \text{ m s}^{-2}$	5.374

Interpretive Boundary

These absolute values are equivalent source scales obtained by applying a Newtonian formula to a uniform static sphere. An expanding FLRW universe has no center or edge and includes pressure and dark-energy terms, so the values must not be read as actual forces or decelerations measured at the end of the universe. By contrast, the density ratio 5.374 is a direct ratio of the present-universe toy model: the radius cancels when both components are evaluated in the same region and geometry.

Dark energy Ω_Λ is not included in the two attractive components above. Because dark energy acts as an expansion term that includes pressure in a homogeneous background, adding it to the simple positive mass ratio of g_M and g_T would mix signs and physical meanings.

9. General Relativity and the Avoidance of Double Counting

In general relativity, the gravity of matter is already represented by spacetime geometry. Calculating both g_M and g_T as the same curvature and then adding them would double count the effect. The present decomposition is effective weak-field bookkeeping. A covariant theory must choose one of the following formulations.

- Source formulation: set $G_{\mu\nu}=(8\pi G/c^4)(T_{\mu\nu}^b+T_{\mu\nu}^T)$ and define $T_{\mu\nu}^T$ as a conserved effective twist stress.
- Response formulation: retain only the baryonic $T_{\mu\nu}^b$ and obtain $v(y)$ by modifying the gravitational action or the response of the field equations.

The two descriptions may be observationally equivalent in the same limit, but they must not be used simultaneously in one calculation. Either $\nabla_\mu(T_b^{\mu\nu}+T_T^{\mu\nu})=0$ or the corresponding conservation law from a modified action is required.

This selection rule applies the B_Q source branch, the B_C geometric branch, and the no-double-counting condition of the earlier horizon study to the present weak-field model [2]. The g_T used here must be the output of one selected branch. The same effect must not be added once as effective stress and again as modified geometry.

$$G_{\mu\nu} = (8\pi G/c^4)(T_{\mu\nu}^b + T_{\mu\nu}^T)$$

one possible covariant completion; presently a hypothesis

(22)

10. SPARC Reality Consistency Test

After fixing a_T from Eq. (2) without refitting it to SPARC, the deep-low-acceleration relation $g_{\text{pred}}=\sqrt{a_Tg_b}$ was tested. The test used 201 of the 2,693 points in the official RAR data with $g_b/a_T<0.03$.

Test	Sample	Mean residual	RMS	Assessment
WRRR deep-low-acceleration relation	201 points	+0.0103 dex	0.1683 dex	2.4% mean offset
Baryonic Newtonian relation	same 201 points	--	0.8766 dex	substantially larger residual
SPARC diagnostic fit for a_T	201 points	$1.20073\times10^{-10}\text{ m s}^{-2}$	--	0.75% from the WRRR value

Applying the external empirical transition function $g=g_b/[1-\exp(-\sqrt{g_b/a_T})]$ to all 2,693 points gives a mean residual of -0.0026 dex and an RMS of 0.1329 dex. This transition function has not yet been derived from a WRRR action, so the result is an external support test rather than a WRRR-specific prediction.

In an auxiliary test, the raw mass models of individual SPARC galaxies were tied to a standard stellar mass-to-light ratio and subjected to quality, inclination, and error cuts. Of 22 galaxies with at least three points in the deep-low-acceleration regime, 59.1% had galaxy-mean residuals within ±0.1 dex and 81.8% within ±0.2 dex. This supports possible universality, but it is not a final fit of the full covariance among distance, inclination, and mass-to-light ratio for each galaxy.

Current assessment

The model has useful reality consistency because its cosmological closure relation reproduces the known galaxy-acceleration scale. The precise assessment is: first reality-consistency test passed; galaxy-by-galaxy universality remains unresolved. Twist stress is a viable causal candidate, but this numerical agreement alone does not establish it.

11. Validation Chain and Falsification Conditions

Verified input	WRRR-specific transformation	Output	Falsification condition	Current conclusion
c, H_0, f_T	$a_T=cH_0v/(f_T/8)$	$1.1918\times10^{-10}\text{ m s}^{-2}$	persistent disagreement with the galactic transition scale	first pass

Verified input	WRRRA-specific transformation	Output	Falsification condition	Current conclusion
M, a_T	$r_T = \sqrt{GM/a_T}$	mass-dependent transition radius	fixed r_T disagrees with rotation-curve transition locations	test required
g_M, v	spatial response in Eq. (9)	rotation field and ρ_T^{eff}	galaxy-specific readjustment of v or a_T is required	unresolved
Ω_b, Ω_T	density ratio in the same region	$g_T/g_M = 5.374$	component correspondence fails in cosmological fits	normalization output
same Φ	common coupling to light and matter	gravitational-lensing map	dynamical and lensing masses disagree	key unresolved test
$H(z), f_T(z)$	$a_T(z) = cH(z)\sqrt{[f_T(z)/8]}$	redshift evolution of the RAR	disagreement with observed evolution after calibration is frozen	prediction candidate

Minimum Dynamical Conditions for the Dark Matter Equivalent

Additional gravity alone does not make twist stress a dark-matter candidate. Applying the conditions formulated in the earlier horizon study, T_T must cluster as an approximately pressureless cold component in cosmic structure formation rather than dilute or free-stream like radiation [2].

$$w_T = P_T / (\rho_T c^2) \approx 0, \quad c_{s,T}^2 = \delta P_T / \delta \rho_T \approx 0$$

$$K_{IJ} > 0, \quad c_{s,i}^2 \geq 0, \quad \tau_{\text{pathological}} > \tau_{\text{obs}}$$

The first equation constrains the background pressure and effective sound speed. The second gives minimum conditions for excluding ghost modes and pathological gradient instabilities. If a covariant action expanded to quadratic order fails these conditions, the dark-matter interpretation of twist stress fails even if it fits galaxy rotation curves.

12. Observational Validation Program

The following sequence develops the hypothesis into a physical theory. The critical rule is that a_T and v must not be retuned at each stage.

- Galaxy rotation curves: calculate complete curves for individual SPARC galaxies with one $v(y)$ and a fixed a_T .
- Baryonic Tully-Fisher relation: test the normalization and scatter of $V_f^4 = G M a_T$ using values fixed in advance.
- Weak and strong lensing: test whether the same Φ inferred from dynamics reproduces light deflection.
- Galaxy clusters: calculate whether residuals not explained by low acceleration arise from the nonlocal structure of spatial stress.
- Solar system and binaries: constrain whether χ_T approaches zero fast enough at $y \gg 1$ to satisfy precision gravity tests.
- Redshift evolution: freeze $a_T(z) = cH(z)\sqrt{[f_T(z)/8]}$ and compare it with the high-redshift RAR.
- Cosmic structure growth: treat ρ_T as a perturbable stress field rather than a background density alone, and test the CMB, BAO, and growth rate together.

The strongest falsifier is a response that fits galaxy rotation but fails for lensing or cosmic structure. In that case, twist stress as the causal interpretation must be rejected or revised at the core level.

13. Separation of the Present Regime from Terminal Twist

The present dark component is modeled as twist stress distributed through three-dimensional space. String-like zero is a terminal-limit hypothesis in which the universe has twisted for so long that no transverse phenotype remains. The present dark-matter sector is not identified in its entirety with that extreme configuration.

The minimum Renderer unit is adopted as a boundary input of the present toy model from the one-bit-per-horizon-area calibration in the earlier horizon study [2]. It is not an independent prediction obtained from two macroscopic cosmic lengths. It remains a conditional calibration fixed by the area law and bit-cell geometry.

$$a_0 = 4\ell_P^2 \ln 2 = 7.243 \times 10^{-70} \text{ m}^2, \quad \ell_0 = \sqrt{a_0} = 2\sqrt{(\ln 2)} \ell_P = 2.691 \times 10^{-35} \text{ m}$$

$$\tau_0 = \ell_0 / c = 8.977 \times 10^{-44} \text{ s}$$

These values are fixed inputs representing a terminal compression or output boundary. The present twist angle and the remaining angle to the maximum cannot yet be calculated because no function $A_\perp(\mathcal{W})$ connects the current background stress to a terminal twist angle. For the same reason, $\mathcal{W}_0=1.26095$ is not converted directly into a physical angle.

Separation Principle

Current tests can constrain a_T , ρ_T , rotation fields, lensing, and cosmic component ratios. The maximum allowed twist of the terminal string-like state remains undetermined until a separate constitutive relation is defined. The minimum area is not inserted artificially into the present stress model to manufacture an angle.

14. Discussion

The main strength of the model is that one numerical closure relation connects the galactic scale a_T to the cosmic quantities H_0 and f_T . Calibrating the factor $1/8$ to the known SPARC scale is part of the model's research status, not a hidden weakness. After calibration, the relation must be frozen and tested against other observations.

The two-dimensional twist interpretation refines the model from an effect that turns on at large mass to an effect revealed by a broad coherent area and low surface density. This direction is consistent with low-surface-brightness galaxies showing strong mass discrepancies despite modest total mass. The interpretation remains qualitative until the surface-density law of the two-dimensional stress is fixed.

Disk flattening and twist stress may influence one another. A theory must define the direction and time evolution of π_{ij}^T , then use the stress inferred from the rotation curve to predict both vertical disk thickness and lensing flattening. This cross-test can determine whether twist is merely a consequence of the disk or also participates in its formation.

The largest theoretical gap is a dynamical derivation of χ_T or ν . A free function can fit almost any rotation curve after the fact and therefore has weak falsifiability. The theory must define an action connecting twist strain, stress, and energy density, and it must fix the high-acceleration suppression rate and nonlocality.

The cosmic mean ratio 5.37:1 translates the standard cosmological dark-matter-to-baryon ratio into WRRRA language. The ratio itself is therefore not an independent prediction. If the spatial-stress interpretation is correct, however, a stress field using the same ratio must connect local galaxy phenomena with lensing and structure growth. That cross-scale consistency is the central WRRRA-specific test.

Conclusion

This study decomposes observed gravity as $g = g_M + g_T$ and defines g_T as the gravitational phenotype of stress stored in spatial twist and deformation. The decomposition is weak-field computational bookkeeping that distinguishes a direct material source from a spatial response. It does not by itself declare a second force competing with general relativity.

The deep-low-acceleration MOND relation can be read as the spatial constitutive response $\chi_T(y) \rightarrow y^{-1/2} - 1$. This interpretation preserves MOND's observational success while supplying a WRRRA causal candidate for its phenomenology. The transition radius grows as $r_T \propto \sqrt{M}$, but the relative effect is controlled by local acceleration and surface density.

If spatial twist is a two-dimensional deformation stress, the transition scale a_T can be read as a surface-density threshold of $\Sigma_* \approx 272 \text{ M}_\odot \text{ pc}^{-2}$ in the spherical approximation and about $136 \text{ M}_\odot \text{ pc}^{-2}$ in the disk normalization. The effect is therefore controlled by a sufficiently broad coherent area and a low direct load per unit area, rather than by large total mass alone.

The present-universe calibration gives $a_T = 1.1918 \times 10^{-10} \text{ m s}^{-2}$, about 0.75% from the diagnostic value obtained from the deep-low-acceleration SPARC data. Since $\Omega_T/\Omega_b = 5.374$, the cosmic mean equivalent source ratio of twist stress to direct baryonic gravity is about 5.37:1. The shares within the two attractive components are 84.3% and 15.7%, respectively.

The present conclusion is specific. The model passes a first reality-consistency test connecting known cosmic and galactic quantities and warrants further study. Twist stress as the physical cause remains a hypothesis. Its explanatory force will be established only if one fixed response function passes galaxy-by-galaxy rotation, lensing, cluster, redshift-evolution, and cosmic-structure-growth tests together.

Appendix A Numerical Reproduction Ledger

Calculation	Expression	Result
Hubble rate in SI units	$67.4 \times 10^3 / 3.0856776 \times 10^{22}$	$2.1843 \times 10^{-18} \text{ s}^{-1}$
Critical density	$3H_0^2/(8\pi G)$	$8.5329 \times 10^{-27} \text{ kg m}^{-3}$
Baryon parameter	$\Omega_b=(\Omega_{bh^2})/h^2$	0.049309
Baryon density	Ω_{bp_c}	$4.2075 \times 10^{-28} \text{ kg m}^{-3}$
Twist density	$0.265\rho_c$	$2.2612 \times 10^{-27} \text{ kg m}^{-3}$
Density ratio	$0.265/0.049309$	5.37425
Observable-universe radius	46.3 Gly	$4.3803 \times 10^{26} \text{ m}$
Equivalent direct acceleration	$(4\pi G/3)\rho_bR$	$5.1526 \times 10^{-11} \text{ m s}^{-2}$
Equivalent twist acceleration	$(4\pi G/3)\rho_TR$	$2.7691 \times 10^{-10} \text{ m s}^{-2}$

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